

Button Basis Trade Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Immeas

ChainDefenders (0x539 & PeterSR)

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Button on Button Basis Trade. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated September 25, 2025, was delivered to Button, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Button Basis Trade code provided by Button. The findings consist of 1 High, 1 Medium & 7 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Button Basis Trade
Repository	button-protocol
Commit	9002f2b0d05b...
Fix Commit	a349cf2c58fe...
Audit Timeline	Sep 15th - Sep 19th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	1	1	0
Low Risk	7	6	1
Informational	10	7	3
Gas Optimizations	1	1	0
Total	20	16	4

6 Audit Scope

```
src/libraries/CoreWriterEncoder.sol
src/BaseTailor.sol
src/BasisTradeTailor.sol
src/BasisTradeVault.sol
src/Pocket.sol
src/PocketFactory.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

BasisTrade is a HyperEVM-integrated asset router built around an ERC-4626 vault. Users deposit a base asset (e.g., USDC) into `BasisTradeVault`, which mints shares (with optional deposit and withdrawal fees) and tracks withdrawal requests via a queue. Funds intended for trading are pushed from the vault to a per-vault `Pocket` wallet through `BasisTradeTailor`. The Tailor then interacts with HyperLiquid HyperCore (encoded by `CoreWriterEncoder`) to add API wallets, move tokens to and from the Core spot wallet, and transfer USD class balances between spot and perp. Withdrawals from the vault are two-step: users request redemption (shares are escrowed and assets "promised"), then the vault is refilled from Core via the Tailor and the queue is processed to pay users from the vault.

Key components:

- **BasisTradeVault (ERC-4626):** manages deposits, fees, a withdrawal queue with share escrow, and oracle-based AUM (`totalAssets`). Creates its dedicated Pocket via the Tailor.
- **BasisTradeTailor:** orchestrates Pocket creation (via `PocketFactory`), maps each Pocket to a "user" (the vault in this setup), and executes HyperCore actions by calling the CoreWriter through the Pocket (`Pocket::exec`). Supports adding API wallets, spot token transfers to and from Core, and USD class transfers between spot and perp. Tracks an aggregate withdrawal intent per Pocket.
- **Pocket and PocketFactory:** Pocket is a minimal smart wallet (deployed as clones) that can `transfer`, `approve` and `exec` calls (no ETH). `PocketFactory` creates Pockets for Tailors and records factory-created Pockets.

9 Attestation

Cyfrin attests that it performed a security review of the Button Basis Trade code provided by Button.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-09-25-cyfrin-button-basis-trade-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Button has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=a1b8d505ff8be1d0308412318564a227d0e11f211d6dc707dda016c3e6d7efad
F=2025-09-25-cyfrin-button-basis-trade-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.